NetApp Keystone Flex Subscription

NetApp Keystone

NetApp
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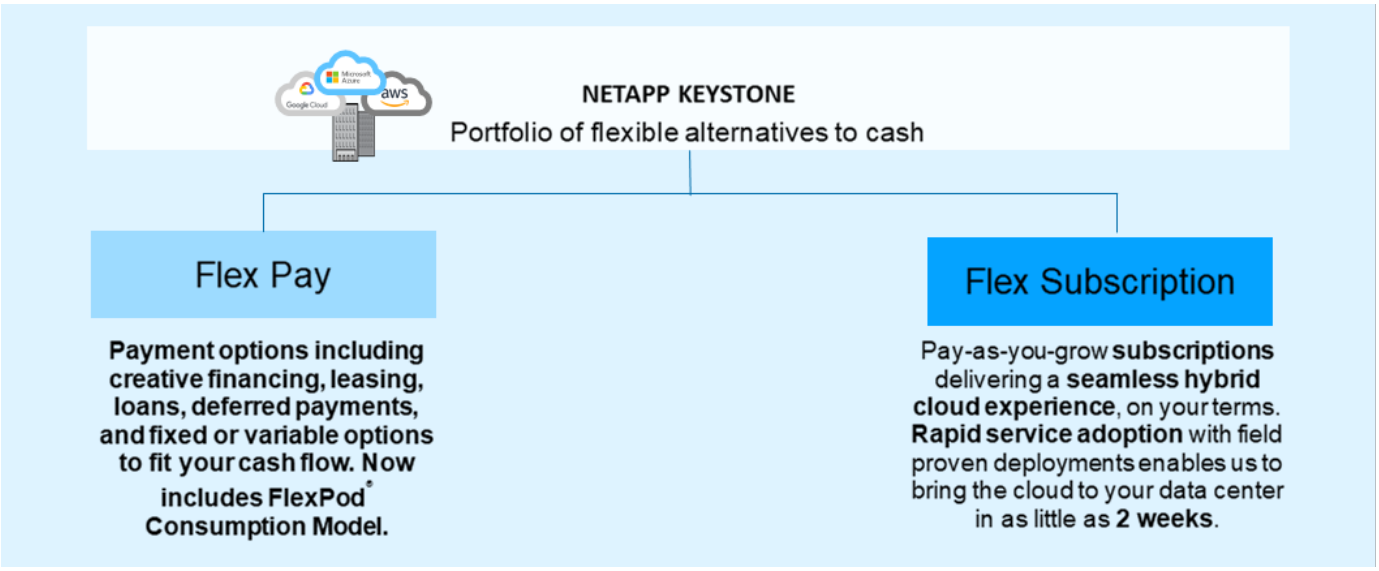
Documentation for NetApp Keystone Flex Subscription

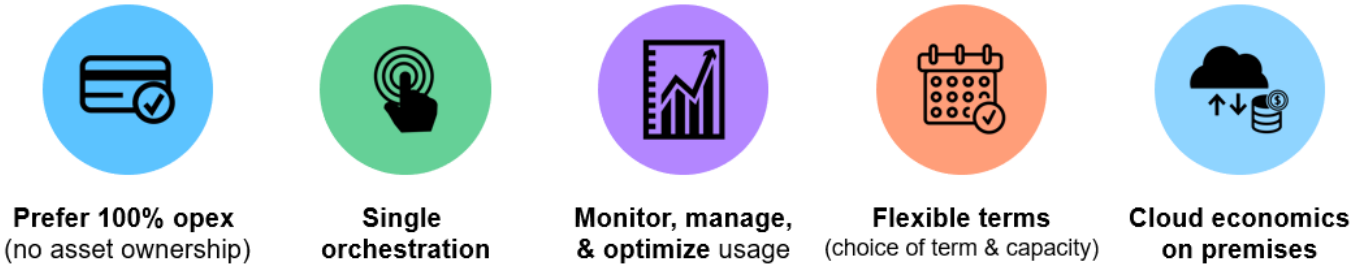
NetApp Keystone Flex Subscription is a part of the NetApp Keystone service offering. NetApp Keystone is a portfolio of on-premises capital expenditure (capex) alternatives to enable the on-ramp to cloud journey, which includes NetApp Keystone Flex Pay and NetApp Keystone Flex Subscription services.

- **NetApp Keystone Flex Pay** - NetApp Keystone Flex Pay (Flex Pay) is a portfolio of flexible financial alternatives that include traditional financing, leasing, and fixed/variable options to meet your customer's cash flow needs.
- **NetApp Keystone Flex Subscription** - NetApp Keystone Flex Subscription (Flex Subscription) is pay-as-you-grow, outcome-based subscription service that brings an on-premises cloud-like experience.



Flex Subscription

Flex Subscription is an outcome-based, on-premises, subscription service model. It enables customers to accelerate time to value by reducing the hurdles related to managing the unpredictable capacity growth and going through the complex procurement cycles. Flex Subscription allows customers to align economics and operations to their business priorities.



Flex Subscription is a subscription-based service offering that provides storage capacity, at predefined performance service levels, for block, file, and object data types that can be deployed on-premises and can be operated by NetApp, a partner, or the customer. In addition to the base service that Flex Subscription provides, add-on services can be chosen at an additional cost. These services are described in the below figure.

- **Data Protection Basic** service provides NetApp SnapMirror and NetApp SnapVault capabilities with default settings.
- **Data Protection Advanced** provides a service with SLAs of RPO=0 by leveraging the NetApp MetroCluster capabilities.

> Flex Subscription uses SnapVault technology for backup operations and SnapMirror for disaster recovery to replicate the data remotely; it does not automatically fail over and restore applications.

Base package

All Flex Subscription services include:

- FlexVol®, FlexGroup®, protocols, clone, and encryption capabilities by default
- Snapshot™ default: 4 hours, 7 days retention
- Free SnapMirror® for initial ingest, provided Premium Bundle on source system

DP Basic*
(for additional fee)

Data protection Basic Services subscription

- Includes SnapVault® and SnapMirror
- Requires additional storage subscription to store replicated data at the target; target can be already purchased and owned NetApp storage
- Disaster recovery snapshot every hour; replicated every 4 hours; retained 7 days
- Backup with SnapVault (snapshot every 4 hours; replicated every 24 hours; retained 7 days)

DP Adv.*
(for additional fee)

Data protection Advanced Services subscription

- MetroCluster™ / RPO 0 Service
- Pre-requisite – data protection Basic

* Available only for file and block data services.

Benefits of Flex Subscription

Flex Subscription provides the following benefits:

- Frees up IT staff from complicated storage-related tasks and allows them to focus on application management
- Reduces upfront capital investment
- Allows customers to meet their demands without overprovisioning
- Aligns data storage costs with business needs/activity
- Simplifies infrastructure provisioning by bypassing complex organizational procurement procedures
- Keeps data secure on their premises
- Enables proper control over compliance, performance, and security

Align costs and usage
Reduce upfront cash and pay only for what you use, avoiding overbuying and overprovisioning

Free IT staff
Free up IT to focus more on innovation and business priorities and less on typical storage tasks

Unlock the best of both clouds
Scale across clouds easily and leverage the public cloud for bursting, data migrations, DR, backup, and tiering

Meet regulatory requirements
Gain improved performance, data protection, compliance, and security with certified and field-proven deployments and services

Flex Subscription service terms and descriptions

- NetApp Keystone Flex Subscription (Flex subscription) is available for a minimum of one year and up to three years. After the initial term, the service is renewable on an annual basis. Capacity can be increased in increments as small as 1 TiB.
- The minimum capacity is 100 TiB per site, and each site can have one or more clusters to meet the minimum capacity requirement. In a partner-operated model, subscriptions with flexible minimums are created for a customer, per site and across service levels.
- The 100 TiB capacity can be one single performance level or a combination of levels.
- Tenant subscriptions are limited to service levels that partners are subscribed to.
- 20% of burst capacity is available at each site; any burst usage is billed only for that billing period. If you need an additional burst requirements that is greater than 20%, contact support.
- Committed capacity or performance levels cannot be altered during a contract term.
- Increasing capacity or changing to higher performance level during term is allowed; however, moving from a higher performance level to a lower level is not permitted.
- Any change request in the last 90 days of the term requires the customer to renew the service for a minimum of one year.

Flex Subscription service capacity definitions

The NetApp Keystone Flex Subscription (Flex Subscription) service capacities include:

- **Logical capacity.** Data placed into the Flex Subscription infrastructure by a customer:
  - All Flex Subscription capacities refer to a logical capacity.

    For example, if a 1 TiB file is stored on the Flex Subscription infrastructure, then at least 1 TiB of capacity must be purchased.
- **Committed capacity.** Minimum logical capacity billed each month for the duration of the term:
  - Capacity is committed to each performance level.
  - Committed capacity cannot be decreased during the term.
  - Committed capacity and additional performance levels can be added during the term.
- **Burst capacity.** Logical capacity, in excess of committed capacity:

- Flex Subscription services provides 20% capacity in excess of the committed capacity.
- Burst capacity can be consumed on an elastic basis and is charged on a daily, average-consumed basis.
- Burst capacity up to 20% is charged at a same rate as the committed capacity.
- Burst capacity greater than 20% of committed is charged at a premium rate. Contact support for any additional burst requirements greater than 20%.

- **Consumed/provisioned capacity.** Sum of all configured volume sizes:
  - Capacity is provisioned through the creation, modification, deletion, or potential auto-growth of volumes.
  - Includes snapshots and clones.
  - The amount of data stored within provisioned capacity is not considered.
  - The amount of data actually written to disk is not considered.
- **Billed capacity.** Monthly bill = (committed capacity [TiB] * committed rate [$/TiB]) + (daily average provisioned burst capacity [TiB] * burst rate [$/TiB]):
  - The monthly bill contains a minimum charge based on the committed capacity.

    The monthly bill varies beyond the minimum charge based on daily average burst capacity consumption.

Performance Service Levels

NetApp Keystone Flex Subscription (Flex Subscription) offers capacity at predefined performance levels.

Each Performance Service Level is defined by its I/O density, which is the ratio of performance (input/output operations per second [IOPS]) and used storage (TiB of stored data) which is IOPS/TiB.

Each volume managed by the Flex Subscription services is associated with a performance service level. All I/O operations and all used storage used on the respective volume are factored into the volume's I/O density calculation.

The below figure defines the performance service levels.

	File and Block						Object	Block		
	Extreme	Extreme w/data tiering[2]	Premium	Premium w/data tiering[2]	Standard	Value	Object	Extreme	Premium	Standard
Workload type	Analytics, databases		VDI, virtualization apps, SW dev		File shares, web servers	Backup	Media repository, archiving	HPC	Video surveillance	Backup
Target IOPS/TiB	6,144	6,144[1]	2,048	2,048[1]	128	N/A	N/A	N/A	N/A	N/A
Max IOPS/TiB	12,288	12,288[1]	4,096	4,096[1]	512	N/A	N/A	5,500	4,000	N/A
Max throughput MBps (32KB/IOP)	384	384[1]	128	128[1]	16	N/A	N/A	43	31	N/A
Latency	<1 ms	<1 ms[1]	< 2 ms	< 2 ms[1]	<17 ms	N/A	N/A	< 0.5 ms	<0.5 ms	N/A
Minimum capacity	100TiB*^						1 PiB	100 TiB	100 TiB	300 TiB
Protocols	NFS, CIFS, iSCSI, FC						S3	FC, iSCSI		

* Minimum one-year term and 100TiB of storage.
^Channel led opportunities: Starts with 15TiB for Extreme

[1] Performance SLAs are applicable for data in hot tier
[2] Max allowed data in hot tier: 25%

I/O density calculations at the volume level are reported to show peak I/O density during the prior week. The peak performance is determined on an hourly time interval. I/O density reports by volume are generated monthly to gauge adherence to the respective service levels.

Service Level metrics and definitions

The following terms and definitions are used within the NetApp Keystone Flex Subscription (Flex subscription) service:

- **GiB, TiB, and PiB.** Measurements of data storage capacity using base of 1024 (1 GiB = 1024^3 bytes, 1 TiB = 1024^4 bytes, and 1PiB = 1024^5 bytes).
- **IOPS/TiB.** The protocol operations per second requested by the application divided by the allocated logical size of the volume.
- **Availability** is measured as a percentage of number of I/O requests successfully responded to by the service, divided by total number of I/O requests made of the service, measured at the service demarcation, in a given month, not including scheduled service downtime or unavailability of required facilities, network or other services to be provided by customer.
- **Durability** is the percentage of data accessed without loss of fidelity, excluding customer-caused deletion or corruption.
- **Target IOPS per TiB.** The guaranteed IOPS for all I/O requests made to a volume before the target IOPs per TiB threshold is reached. Performance on the volume is capped at the selected IOPS per TiB.

> The target IOPS per TiB performance metric is calculated based on the logical consumed capacity in TiB.

- **Latency.** Time to service an I/O request received from a client, measured at the service demarcation (storage controller I/O port).

Data Protection

NetApp Keystone Flex Subscription (Flex Subscription) data protection service can back up your data and is able to recover it if required. The available data protection services

are:

- Snapshots of disks and shares
- Backups of disks and shares (requires data protection service as part of the subscription)
- Disaster recovery for disks and shares (requires data protection service as a part of the subscription)

Backup and disaster recovery services are available as add-on services, while Snapshot is available as a part of the basic storage service.

	Single Region Snapshots (Available as a part of the basic storage service)	**Multi-region Backup (data protection add-on)**	**Multi-region Disaster Recovery (data protection add-on)**
Use case	Mitigate the risk of user or application data deletion or corruption, not against infrastructure loss or failure	Mitigate the risk of complete loss of data on the primary volume due to infrastructure loss or failure	Mitigate the risk of complete loss of data on the primary volume due to infrastructure loss or failure with a recovery time objective
Policy	Hourly, daily, weekly, and monthly	Number of backups to retain based on hourly, daily, weekly, and monthly Snapshots	1 hour, 4 hours, and daily
Topology	Source only	Backup	Async replication target
Target replication service level[1]	n/a	Standard	Same as primary

[1]Additional storage capacity to be subscribed

Subscription to a basic Flex Subscription service does not automatically back up your data. You should subscribe to add-on data protection services and configure your system for data backup and disaster recovery services. If your storage system is not managed by Flex Subscription services, NetApp can still support protecting the data on your storage system and help in connecting it with your Flex Subscription services. However, NetApp is not responsible for any backup failures.

Tiering

NetApp Keystone Flex Subscription service includes a tiering capability that identifies less frequently used data and tiers it to a cold storage that is owned, deployed, and managed by NetApp.

The tiering capability leverages the NetApp FabricPool technology that enables automated tiering of data to low-cost object storage tiers either on or off premises. With this capability, infrequently accessed data is automatically tiered to a lower cost storage either on premises or in the cloud, based on the services agreed upon.

Partners and tenants can avail this capability easily by opting for the two preconfigured service levels, the Extreme-tiering and Premium-tiering service levels while provisioning their storage. The Extreme-tiering has

the same QoS policies as the Standard, Extreme, and Premium service levels.

The add-on tiering capability is available only with Extreme and Premium service tiers. NetApp assumes 25% of data is hot and 75% is less frequently used and can be moved to a cold storage. Billing is determined based on the duration per volume is in each service level.

The following features are enabled:

- You can create reports of the inactive data for your disks and file shares and decide upon whether to change the service level. On moving or changing the tiering policy, the latencies can be higher if data is accessed from cold tier.
- You can change the service level of the volumes from Extreme and Premium to Extreme-tiering and Premium tiering respectively, provided that the destination tiering is enabled on the cluster.
- Likewise, you can change the tiering service levels to non-tiering for your volumes.
- Enable and disable backups for a volume on a tiering service level.
- Enable and disable disaster recovery for a volume on a tiering service level.

Operational model, roles, and responsibilities

NetApp Keystone Flex Subscription (Flex Subscription) is based on the model of tenancy.

Flex Subscription offers three operational models for service delivery.

- **NetApp-operated model** allows the customer to subscribe to the offered services (according to the selected performance tiers and storage service types) and selects the NetApp-operated option at an extra cost. NetApp defines the architecture and products, installs at the customer premises, and manages the day-to-day infrastructure management operations by using NetApp storage and IT resources. Available storage service types are file, block, and object. Storage subscriptions based on Cloud Volumes Service for GCP and AWS can also be managed through your NetApp Keystone instance.
- **Partner-operated model** is similar to the NetApp-operated model, but with the partner operating the service for their end customers. In this model, the partner is the referenced contracted party. Tenants are customers of partners or service providers and have no billing relationship with NetApp. A partner-operated model usually has a multi-tenant environment where tenants and end customers/subtenants have their own subscriptions that are billed by the service provider/partner. The partner admin performs the administrative tasks for all the tenants. The functions that an admin can perform in a partner-operated model are different from that of an admin in a NetApp-operated model.
- **Customer-operated model** allows the customer to subscribe to an offered service, according to the selected performance tiers and storage service types. NetApp defines the architecture and products and installs at the customer premises and allows customers to manage the infrastructure using their storage and IT resources. A customer can be tied to NetApp or a partner/service provider, and based on that, the service requests can be raised and addressed to NetApp or the service provider. A customer admin can perform the administrative tasks in a customer-operated environment. These tasks are tied to the tenants and subtenants for the specific customer.

The features and options in the offering vary based on the models. For information, see Roles and operations of service providers and customers

This figure summarizes the overall service lifecycle model and categorizes the associated services.

Roles and Responsibilities across Service Lifecycle

	NetApp Operated	Partner Operated	Customer Operated	
	NetApp	Partner	NetApp	Customer
Build	☑	☑	☑	
• Install	☑	☑	☑	
• Configure	☑	☑	☑	
• Deploy	☑	☑	☑	
• Onboard	☑	☑	☑	
Monitor & Administer	☑	☑		☑
• Monitor	☑	☑		☑
• Alert	☑	☑		☑
• Administer	☑	☑		☑
• Report	☑	☑		☑
Operate & Optimize	☑	☑		☑
• Manage capacity	☑	☑		☑
• Manage performance	☑	☑		☑
• Manage SLA	☑	☑		☑
Support	☑	☑	☑	
• Support customer	☑	☑	☑	
• HW break fix	☑	☑	☑	
• SW support	☑	☑	☑	
• Upgrades and patches	☑	☑	☑	

Roles and responsibilities summary

The following list summarizes NetApp's roles and responsibilities:

- NetApp delivers, installs, configures, and enables the applicable service (including applicable version of NetApp Service Engine at a customer-designated data center or a CoLo). NetApp is responsible for the uninstallation at the end of the contract term or if the customer chooses to terminate the contract sooner.
- While interfacing with other IT service providers, NetApp cooperates with the customer's IT service providers or their technical team.
- Performance of the NetApp-operated services are included, with the assumption that shared responsibilities are applicable to the supported environment.

Tenancy and multi-tenancy in Flex Subscription

NetApp Keystone Flex Subscription (Flex Subscription) uses the concepts of **Tenant** and **Subtenant** as hierarchical entities that own logical storage resources. The concept of multi-tenancy is also supported, where multiple tenants are tied to a partner or service provider. The entities partner and service provider are used interchangeably here.

In the context of Flex Subscription, a single tenancy is a NetApp-operated model, while a multi-tenancy is a partner-operated model.

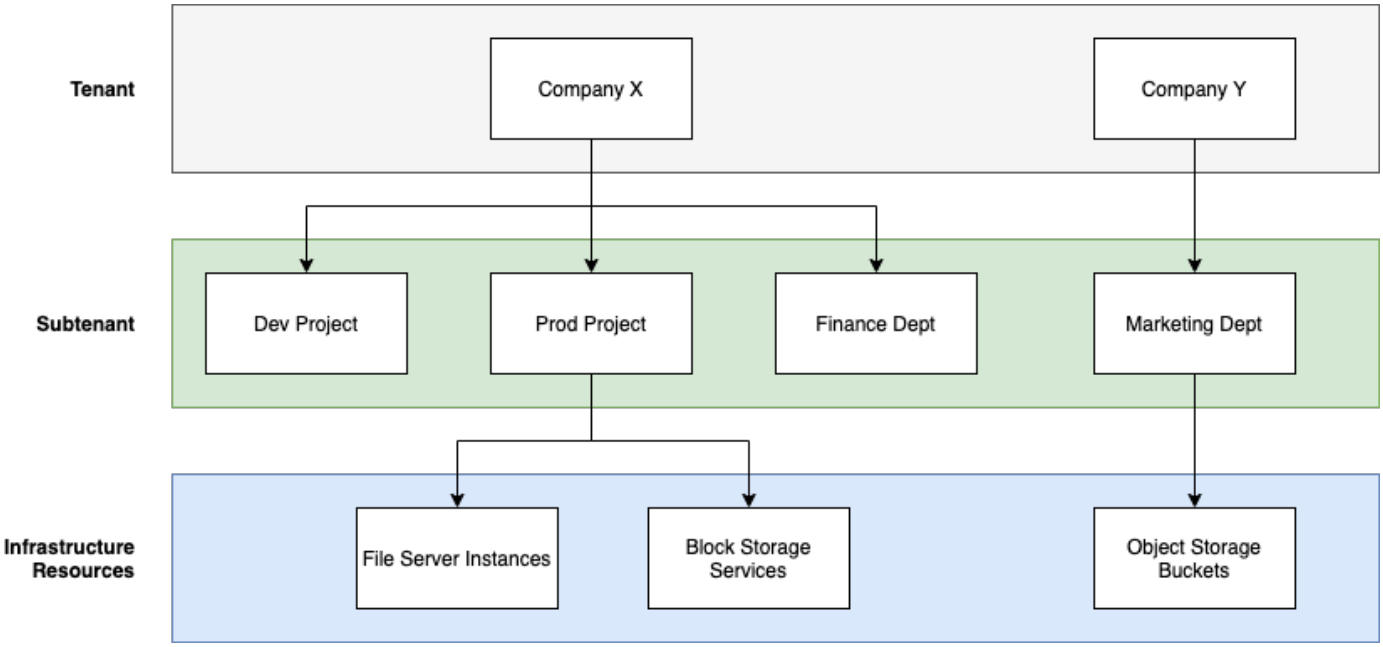
Tenants

A tenant is the primary billing entity defined in Flex Subscription. Each customer that is onboarded into Flex Subscription and holds a subscription to the services exists as a Flex Subscription tenant. The customer provides NetApp with a customer name and/or identifier for the purposes of onboarding tenants and tracking subscription dates and commitment terms.

In a multi-tenancy, a partner is a tenant/customer of NetApp in a Flex Subscription environment. The partner, in turn, can bear multiple tenants or customers, who are linked to end customers/subtenants. In this model, tenants/subtenants are the customers of the service providers, and have no billing relationship with NetApp.

Subtenants

Flex Subscription subtenants exist entirely within a parent tenant with a many-to-one relationship. Subtenants provide logical resource separation within a tenancy and are used as a basis for show-back reporting. All storage resources (that is file services, block storage, and object storage) belong to a subtenant.



Subscription model in a NetApp-operated (single tenancy) environment

- A subscription is created by NetApp admin or GSSC for a tenant and zone.
- The subscription names are derived from the billing entity for which the tenant is subscribed.
- In the subscription, the start-date and contract term are configured.
- There can be only one active subscription for each tenant-zone
- One subscription can have multiple rate plans and each rate plan corresponds to a service level.
- Each rate plan has a committed capacity per service level.
- Service levels can include:
  - Extreme
  - Extreme-Tiering

- Performance
- Performance-Tiering
- Value
- Data protection for each of the service levels
- Advanced Data Protection for Extreme, Performance, Value
- Storage objects

Subscription model in a multi-tenant environment

- Service Providers are Flex Subscription customers and have subscriptions as tenants. The subscriptions are based on:
  - Commitment per service level and zone
  - Charged on allocated capacity with 100TiB minimum
  - Burst charges apply for 100-120% of committed capacity
- NetApp charges the providers monthly, as a part of their usual tenancy terms.
- For a service level to be available to tenants, the service provider or partner should first have a Flex Subscription in place for the service level.
- The service provider creates tenant subscriptions per service level, zone, and flexible minimums.
- Service providers can sell more capacity to their tenants than they have purchased from NetApp (oversubscription). Therefore, the capacity used by tenants is not limited by the capacity the service provider has subscribed to.
- Tenants can use storage capacity over their subscribed amount, that is listed as 'burst' on usage reports.
- Tenant usage reports are available to partners for viewing on a daily or monthly basis.
- Tenants can create subscriptions for longer duration as compared to the corresponding Flex Subscription, but a warning message is displayed to the end customer during that activity.
- Flex Subscription for a partner is configured by NetApp admins or GSSC. Management of Flex Subscription and tenant subscriptions is performed by a user with Partner admin role.
- Users with the tenant admin roles can only view the tenant subscription (not partner’s Flex Subscription). They can update the given subscription to change capacity and service level. They can raise service requests for additional subscriptions.
- The partner admin can create another subscription either when the existing subscription is expired, or for a future date when the existing subscription is no longer valid. The start date for a new subscription must be greater than or equal to current end date.

Flex Subscription infrastructure

This section describes the NetApp Keystone Flex Subscription (Flex Subscription) infrastructure architecture and management application for the NetApp and customer-operated environments.

Flex Subscription infrastructure architecture, design, choice of technology, and component products reside solely with NetApp. NetApp reserves the rights to take the following actions:

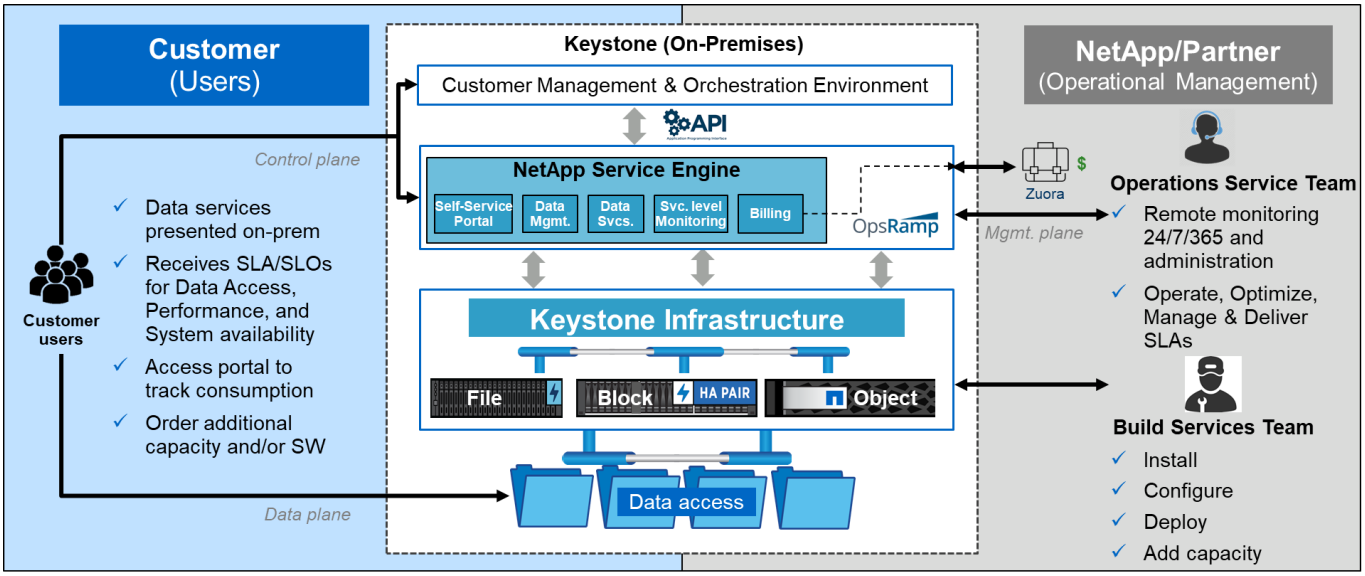
- Select, substitute, or repurpose products.

- Refresh products with new technology when deemed appropriate.
- Increase or decrease capacity of the products to meet service requirements.
- Modify architecture, technology, and/or products to meet service requirements.

The Flex Subscription infrastructure includes multiple components:

- User interface (web portal) of NetApp Service Engine.
- NetApp Service Engine APIs for integration.
- The Flex Subscription infrastructure that includes storage controllers
- Tools to manage and operate the service such as OpsRamp, Active IQ, and Active IQ Unified Manager.

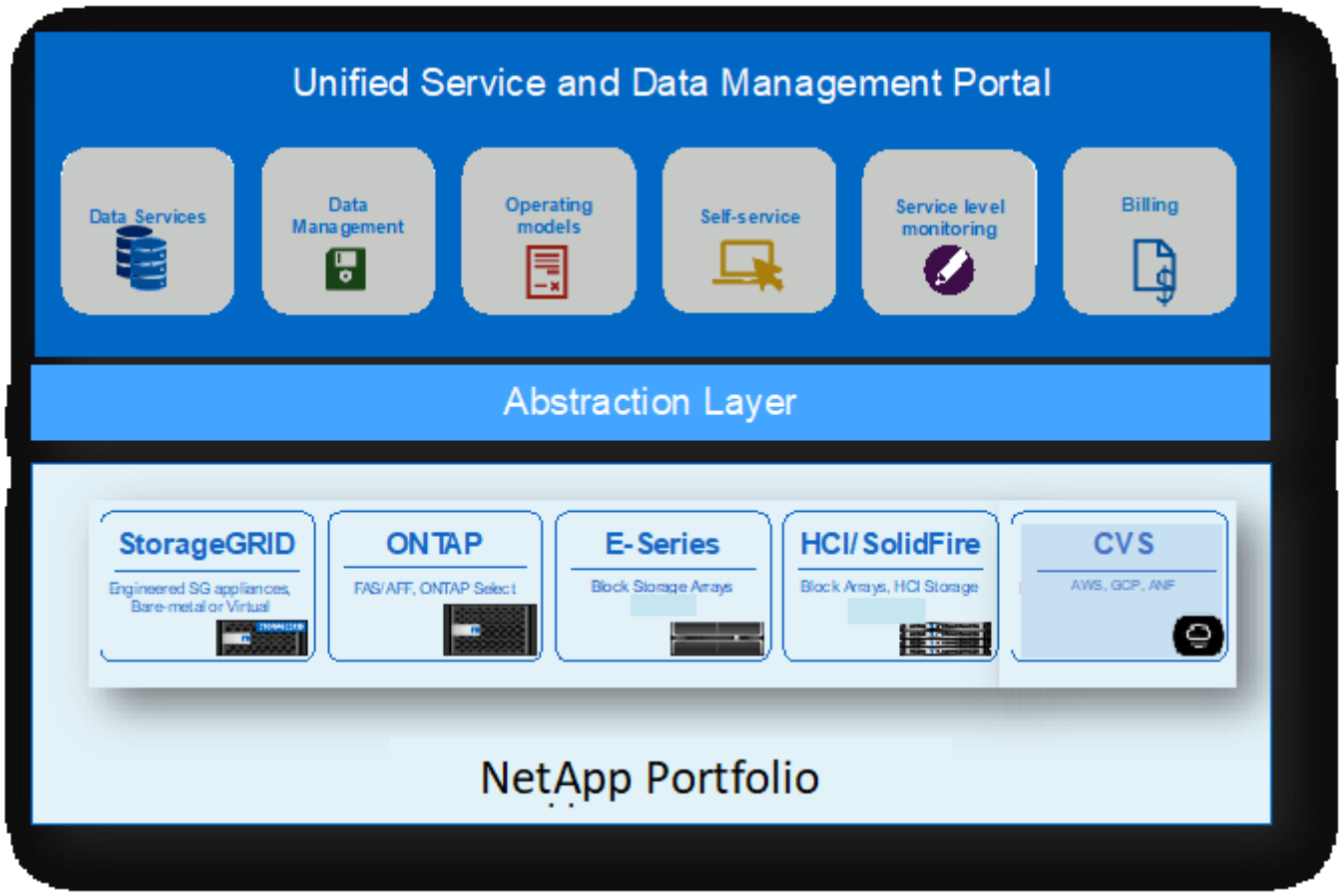
For more information about NetApp Keystone integration with Active IQ, see Active IQ Digital Advisor Documentation



NetApp Service Engine

You can use the NetApp Service Engine web portal to manage and monitor your Flex Subscription service in a NetApp, partner, or customer-operated model. The portal consists of the following features:

- A graphical interface (NetApp Service Engine web interface) that supports monitoring and simple storage provisioning.
- A set of NetApp Service Engine REST APIs that allow more advanced setup and storage management actions.



The NetApp Service Engine portal is a single interface that allows customers to perform the following tasks:

- Subscription dashboard. View status of subscribed capacity and how much is being consumed.
- Storage provisioning. Workflows to create your NFS/CIFS file shares, FC/iSCSI disks, and S3 bucket.
- Data protection. Create snapshots and set backup policy or data replication on the provisioned file share or disk.
- Reports. View file shares and disks inventory and also trend on capacity usage against different service levels.
- Manage subscription. Order additional capacity to an existing subscription or a new service through the portal.
- Submit service requests or support issues and view their status

The complete functionality of NetApp Service Engine is available only in the NetApp-operated model. In a customer or partner-operated model, the NetApp Service Engine functionalities are limited to reporting and subscription management.

Site requirements

There are multiple requirements for enabling NetApp Keystone Flex Subscription (Flex Subscription) services in your environment, such as space, racks, PDUs, power, and cooling, with additional network and security requirements discussed later.

Space

Floor space to host the Flex Subscription infrastructure equipment (to be provided by customers). NetApp provides the weight specifications based on the final configuration.

Racks

Four post racks in the customer-operated offering (to be provided by customers). In the NetApp-operated offering, either NetApp or the customer can provide the racks, depending on requirements. NetApp provides 42 deep racks.

PDUs

You should be providing the power distribution units (PDUs), connected to two separate, protected circuits with sufficient C13 outlets. In the customer-operated offering, in some cases, C19 outlets are required. In the NetApp- operated offering, either NetApp or the customer can provide the PDUs, depending on requirements.

Power

You should provide the required power. NetApp will provide the power requirement specifications based on 200V rating (Typical A, Max A, Typical W, Max W, Power cord type, and quantity), based on the final configuration. All components have redundant power supplies. NetApp will provide the in-cabinet power cords.

Cooling

NetApp can provide the cooling requirement specifications (Typical BTU, Max BTU), based on the final configuration and requirement.

Networking

Depending on customer requirements and the storage controllers used, NetApp can provide 10 Gb, 40 Gb, and 100 Gb connectivity at the customer site.

NetApp provides the required transceivers for NetApp-provided Flex Subscription infrastructure devices only. You should supply transceivers required for customer devices and cabling to the NetApp-provided Flex Subscription infrastructure devices.

Flex Subscription Services Operations

NetApp Keystone Flex Subscription Services are run by leveraging the NetApp Global Services and Support Center (GSSC), NetApp Keystone Success Manager, and NetApp Keystone Service Delivery Manager.

NetApp Global Services and Support Center (GSSC)

NetApp provides operational services remotely to NetApp Keystone customers. These services encompass a range of operational disciplines across storage management activities, including asset and configuration management, capacity and performance management, change management, event, incident and problem management, service request fulfillment, and reporting. NetApp will demonstrate a state of control and supporting evidence available as required.

Additional Information and Support Contact

The NetApp Global Services and Support Center (GSSC) team primarily support the services to NetApp Keystone Flex Subscription customers.

Raise a ticket directly from the NetApp Service Engine portal (**Support > Service Requests**) with the appropriate details of the issue for assistance.

You can also use the following information to reach out to the support team.

- Global Service Contacts :
  https://www.netapp.com/us/contact-us/support.aspx
- If you have an open case/ticket that needs to be escalated, please send an email to one of the following addresses:
  Keystone.services@netapp.com
  Keystone.escalations@netapp.com
- NetApp uses OpsRamp, a cloud-based remote gateway solution to proactively monitor and connect to the NetApp Keystone environment for troubleshooting purposes. For information on OpsRamp, see https://www.opsramp.com/#.

> In a partner-operated model, the tenant and subtenant's service requests are assigned to the partner's service desk. The partner's support tool might have integration with OpsRamp and GSSC applications. Only L3 issues are escalated to NetApp through GSSC.

To learn more about the information that is described in this document, review the following documents and/or websites:

- NetApp Keystone
  https://www.netapp.com/us/solutions/keystone/index.aspx
- NetApp Product Documentation
  https://docs.netapp.com

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